6.2.1 Action Group Control Course

This section aims to guide you on how to obtain the data of miniArm's action group via Arduino platform, as well as how to write and execute the action group.

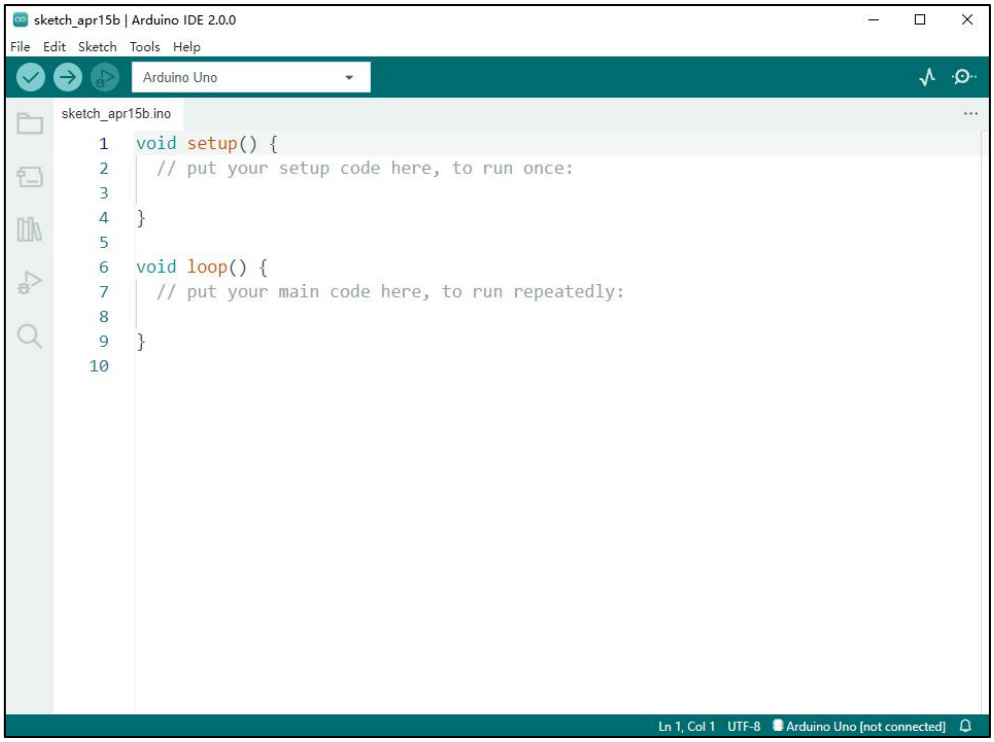
1. Read Action Group

1.1 Basic Program Download

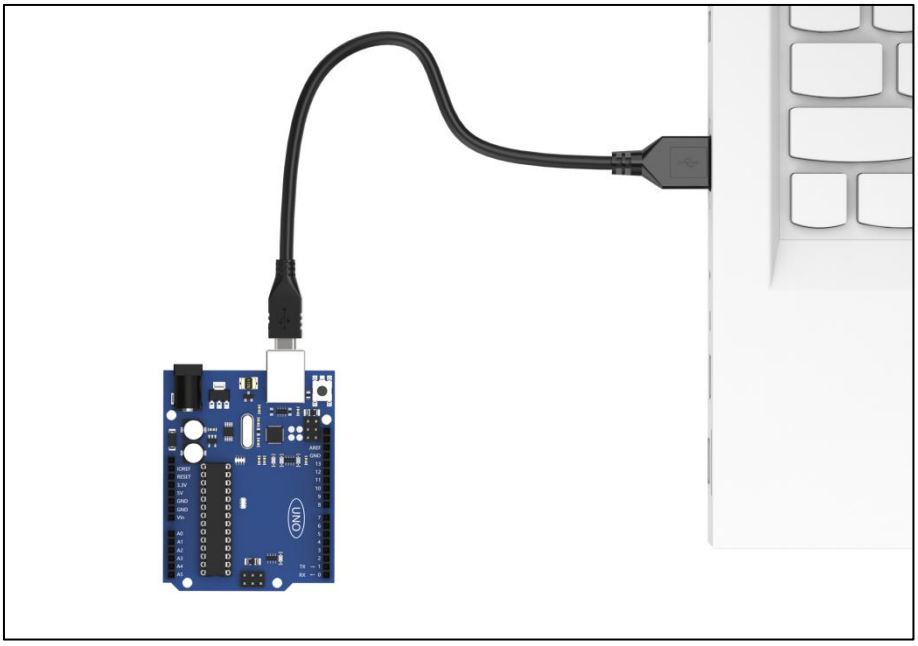
◆ Remove the Bluetooth module before downloading the program. Or the program will fail to download because of the serial port conflict.

◆ Please switch the battery box to "OFF" when connecting Type-B download cable. This action prevents the download cable from accidentally touching the power pins of the expansion board, which may cause a short circuit.

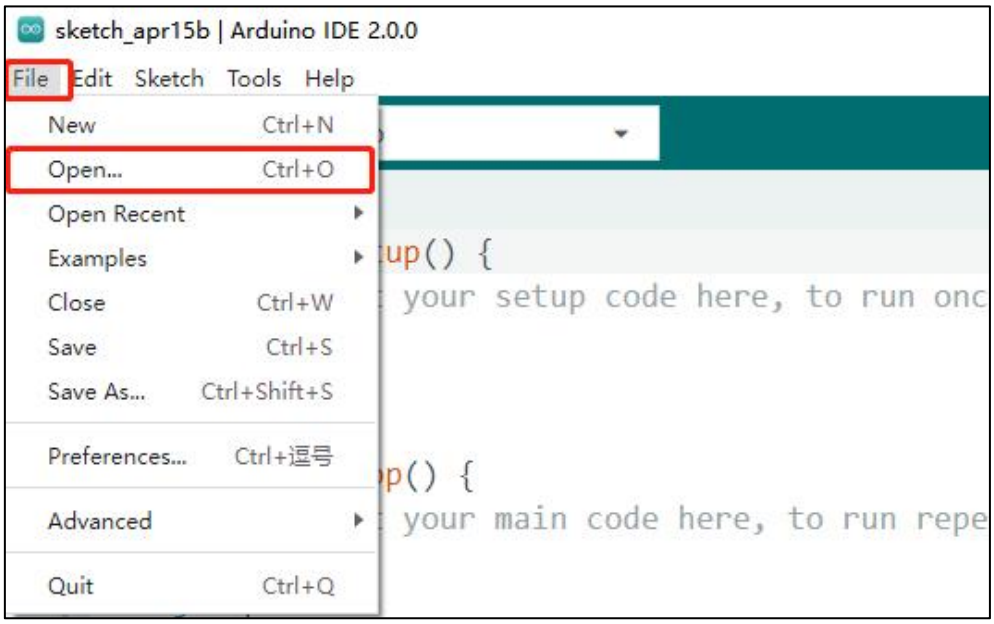
1) Double click to open it.



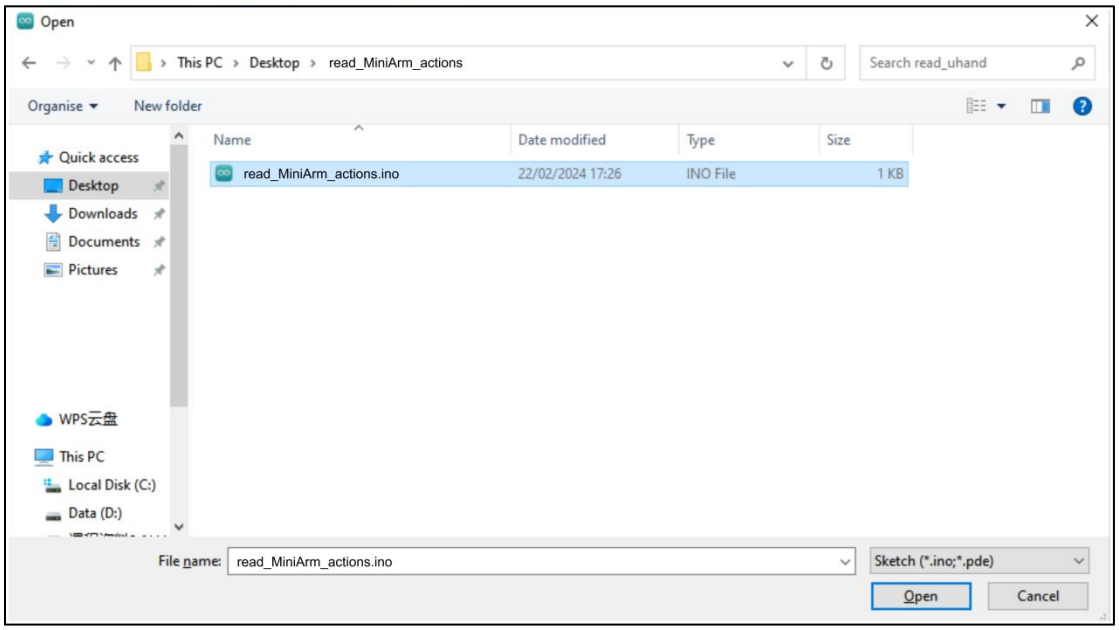
2) Connect the Arduino to the computer with the UNO cable (Type-B).



3) Click “File -> Open”.

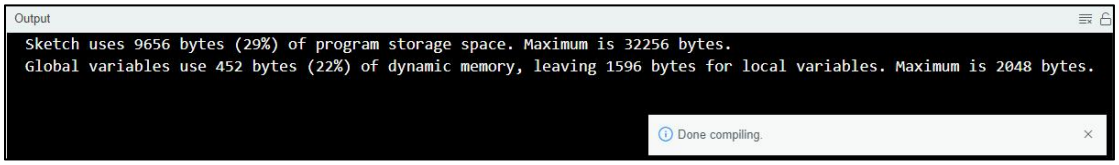


4) Access “6.2.2 Read Action Data Program”. Select “read_mini_actions.ino” program in “read_mini_actions” folder and click “Open”.

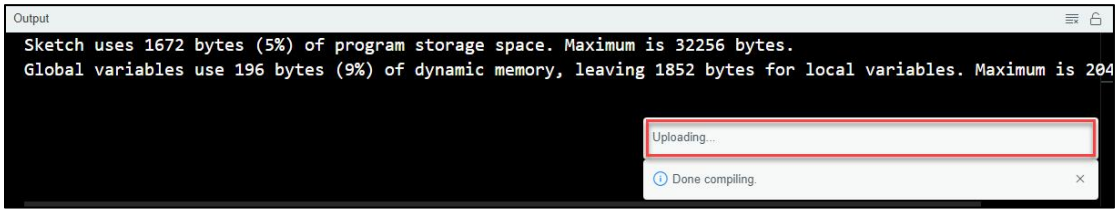


5) Locate Arduino development board in “Select Board”. (Take Arduino Uno as an example. COM ports are not unique, and their numbers can be checked in computer's device management. Take COM6 as an example.)

6) After confirming that everything is correct, click to verify the program. If the program is correct, the status area shows “Compiling Project -> Completed” in turn. Then it displays the number of bytes used by the current project, the program storage space and etc.



7) After successful compilation, click to upload the program to the development board. The status area shows “Compiling Project-> Uploading-> Upload successfully” in turn. After that, the status area stops printing and uploading information.

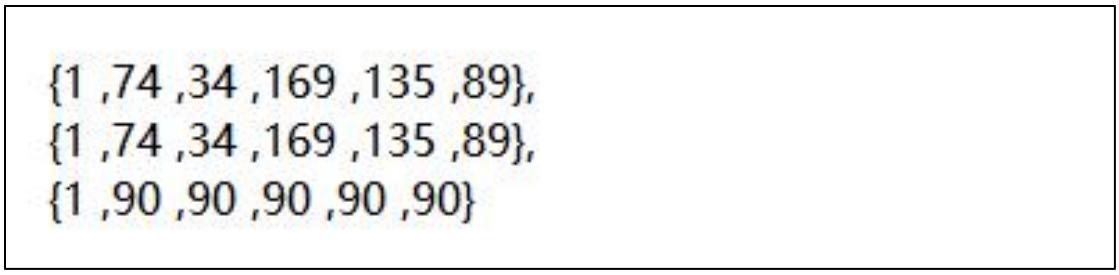


1.2 Get Action Group Data

1) Keep the Type-B cable connected to the computer. Do not unplug it, as it will hinder communication.

2) Press “Ctrl+Shift+M” to open the serial monitor to view real-time servo data. The data corresponds to the angles of servos 1 to 5 respectively. The 0th bit in the data tells whether the current action group is a valid data or not. “1” is a valid data, which is used to determine if the action is successfully executed during action execution.

1) Rotate the knobs S1 to S5 on the miniArm to adjust the current rotation angles of the servos. And then save these actions in series to create a new action group.



2. Run Action Group

2.1 Write Action Group

1) Add the acquired action group data into “6.2.3 Action Running Program\mini_actions_group\actions.h”.

2) The macro definition “action_count” represents the number of current action groups. If a new action group needs to be added, please add 1 to the original data.

```
#include <Arduino.h>
#define action_count 2 //Number of action groups

static uint8_t action[action_count][20][6] =
    {
      //Action Group 1
      {{1 ,90 ,0 ,130 ,125 ,90},{1 ,93 ,0 ,141 ,135 ,0},
      {1 ,93 ,0 ,141 ,135 ,174},{1 ,90 ,90 ,90 ,90 ,90}, {0,0,0,0,0,0}}
      //Action Group 2
      {{1 ,74 ,34 ,169 ,135 ,89},{1 ,74 ,34 ,169 ,135 ,89},
      {1 ,90 ,90 ,90 ,90 ,90}, {0,0,0,0,0,0}},
    };
```

Please make sure that the current action data is a one-dimensional array when adding new action group data. In the action group data, the 0th bit in the data tells whether the current action group is a valid data or not. The 1st to 5th bits correspond to the angles of the servos 1 to 5 respectively.

Note: Each action group data must end with {0,0,0,0,0,0,0,0} to indicate the current action group running has finished.

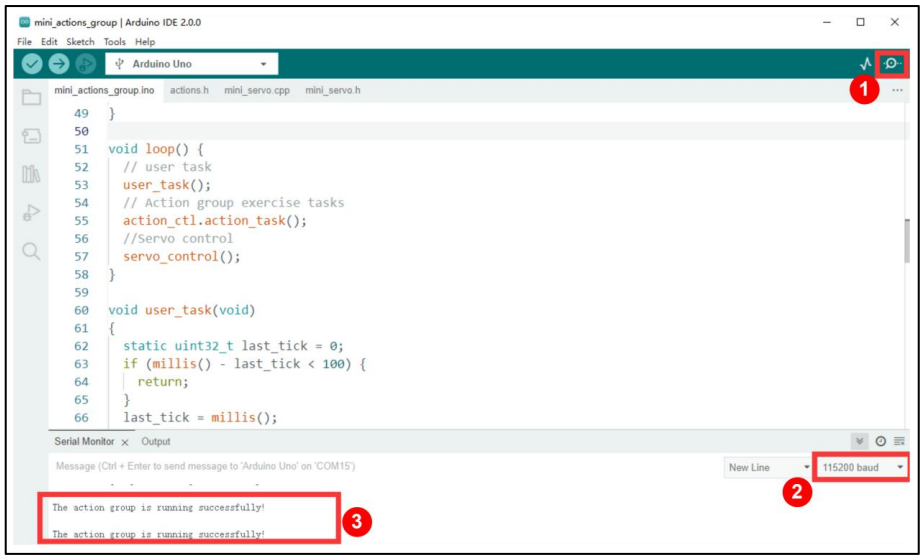
3) After adding the action group data, please refer to “1.1 Basic Program Download" to download the "mini_actions_group.ino" program to the Arduino UNO controller board.

4) Once the download is completed, power on the robotic arm to run the set action group.

2.2 Program Outcome

Power on the robotic arm. Keep the Type-B cable connected to the computer. Turn on the serial monitor, and select a baud rate of “115200”.

Once the action group is completed, the serial port monitor displays “The action group is running successfully!”, indicating successful execution of the action group.



2.3 Program Analysis

1) Initialize the serial port as “115200”, then the servo pins. Call the error reading function “read_servo_offset()” to read the current deviation of the robotic arm.

```
void setup() {
  // put your setup code here, to run once:
  Serial.begin(115200);
  // Set the serial port’s timeout for reading data.
  Serial.setTimeout(500);

  // Assign servos’ IO port
  for (int i = 0; i < 5; ++i) {
    servos[i].attach(servoPins[i]);
  }

  //Call error reading function
  read_servo_offset();

  delay(2000);
  Serial.println("start");
}
```

2) Within the “user-task ()” function, set the sequence number of the currently executing action group by “action_ctl.action_set()” function. Take the code “action_ctl.action_set(1)” as an example. Set the current action group number to be executed as “1”. Get the currently executing action group through the “action_state_get( )” function, whose return value is the sequence number of the currently executing action group. If the execution is complete, it returns “0”.

```
void user_task(void)
{
  static uint32_t last_tick = 0;
  if (millis() - last_tick < 100) {
    return;
  }
  last_tick = millis();

  static uint32_t step = 0;
  switch(step)
  {
    case 0:
      //Action group control
      action_ctl.action_set(2);//Execute Action Group 1
      Serial.print("action run.");
      step = 1;
      break;
    case 1:
      if(action_ctl.action_state_get() == 0)
      {
        Serial.println("");
        Serial.println("The action group is running successfully!");
      }else{
        Serial.print(" .");
      }
      break;
  }
}
```

3) Use the “action_ctl.action_task()” function to set the corresponding servo angles for the current action.

```
void HW_ACTION_CTL::action_task(void){
  static uint32_t last_tick = 0;
  static uint8_t step = 0;
  static uint8_t num = 0 , delay_count = 0;
  if(action_num != 0 && action_num <= action_count){
    // Interval time
    if (millis() - last_tick < 100) {
      return;
    }
    last_tick = millis();
    switch(step){
      case 0: //Running actions
        if(action[action_num-1][num][0] != 0){
          extended_func_angles[0] = action[action_num-1][num][1];
          extended_func_angles[1] = action[action_num-1][num][2];
          extended_func_angles[2] = action[action_num-1][num][3];
          extended_func_angles[3] = action[action_num-1][num][4];
          extended_func_angles[4] = action[action_num-1][num][5];
          step = 1;
        }else{ //If the operation is completed
          num = 0;
          // Clear action group variables
          action_num = 0;
        }
        break;
```

```
        case 1: //Waiting for action to run
          delay_count++;
          if(delay_count > 2){
            num++;
            delay_count = 0;
            step = 0;
          }
          break;
        default:
          step = 0;
          break;
      }
    }
}
```

4) Next, call “servo_control()” to control servos rotating to corresponding angles for achieving related action. Adjust the deviation of the current robotic arm before running the action.

```
void servo_control(void) {
  static uint32_t last_tick = 0;
  if (millis() - last_tick < 20) {
    return;
  }
  last_tick = millis();

  for (int i = 0; i < 5; ++i) {
    servo_expect[i] = action_ctl.extended_func_angles[i] + servo_offset[i];
    if(servo_angles[i] > servo_expect[i])
    {
      servo_angles[i] = servo_angles[i] * 0.9 + servo_expect[i] * 0.1;
      if(servo_angles[i] < servo_expect[i])
        servo_angles[i] = servo_expect[i];
    }else if(servo_angles[i] < servo_expect[i])
    {
      servo_angles[i] = servo_angles[i] * 0.9 + (servo_expect[i] * 0.1 + 1);
      if(servo_angles[i] > servo_expect[i])
        servo_angles[i] = servo_expect[i];
    }

    servo_angles[i] = servo_angles[i] < limt_angles[i][0] ? limt_angles[i][0] : servo_angles[i];
    servo_angles[i] = servo_angles[i] > limt_angles[i][1] ? limt_angles[i][1] : servo_angles[i];
    servos[i].write(i == 0 || i == 5 ? 180 - servo_angles[i] : servo_angles[i]);
  }
}
```

If modifying the rotation speed of the robotic arm’s servos, you should ensure that the sum of actual and desired values is “1”.

For instance, if you modify the actual value as “0.5”, you should change the corresponding desired value to “0.5”. After modifying, download the

program, and will see a significant increase in the rotation speed of the robotic arm's servos.